

Completing the square



1 Halve the coefficient of x and then square it.

2

a No

b 9

3 $\frac{9}{4}$

4

a 25

b 1

c 8

d $\frac{1}{4}$

e $\left(\frac{m}{2}\right)^2$

f $\frac{4}{25}$

5

a $x^2 + 4x + 4 = (x + 2)^2$

b $x^2 - 5x + \frac{25}{4} = \left(x - \frac{5}{2}\right)^2$

c $x^2 - \frac{7}{4}x + \frac{49}{64} = \left(x - \frac{7}{8}\right)^2$

6

a $(x + 6)^2 - 36$

b $(x - 6)^2 - 36$

c $(x + 1)^2 + 1$

d $(x - 7)^2 + 7$

e $(x + 2)^2 - 2$

f $(x - 3)^2 - 4$

g $\left(x + \frac{3}{2}\right)^2 + \frac{15}{4}$

h $\left(x - \frac{7}{2}\right)^2 + \frac{7}{4}$

7

a $3\left(x + \frac{3}{2}\right)^2 + \frac{5}{4}$

b $4\left(x - \frac{11}{8}\right)^2 - \frac{9}{16}$

8 $4\left(\left(x - \frac{9}{2}\right)^2 - \frac{1}{2}\right)$

9

a $(b + 7)^2$

b $(y - 5)^2$



c $(x + 5)^2$

d $(x + 7)^2$

e $(x - 6)^2$

f $(x - 8)^2$

g $(9x - 4)^2$

h $(9x + 2)^2$

i $(7x - 2)^2$

j $(9 + x)^2$

k $(6 - x)^2$

l $(5 + 6r)^2$

m $7(d + 8)^2$

n $4(x + 5)^2$

o $3(x - 4y)^2$

p $-6(r + 8)^2$

q $-3(x - 2)^2$

r $\left(b - \frac{7}{2}\right)^2$

s $\left(x + \frac{3}{2}\right)^2$

t $\left(x - \frac{3}{2}\right)^2$

10

a $y = (x - 3 + \sqrt{2})(x - 3 - \sqrt{2})$

b $y = (x + 4 + \sqrt{2})(x + 4 - \sqrt{2})$